FACTORING A MORPHISM THROUGH A MONOMORPHISM VIA THE OPEN PART AND THE COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof builds the fibre product, shows its projection is etale over the closed part via the infinitesimal criterion and is a monomorphism, identifies it as an open immersion, and concludes by the covering of |X|

Editorial note. The label lemma-factor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the completion functor section: morphisms with image in T factor through X_{/T}-related fibre products (Formal Spaces, Section formal-spaces-section-completion)
- the infinitesimal criterion for etaleness (More on Morphisms of Spaces, Lemma spaces-more-morphisms-lemma-check-smoothness-on-infinitesimal-nbhds)
- an etale monomorphism of algebraic spaces is an open immersion (Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open)

Source reference. Original section 0AIX: sources/stacks-chow/tags.txt, source lines 13280–13280. Prerequisite pointer invoked at proof line(s) 6472; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0APQ: sources/stacks-chow/tags.txt, source lines 13483–13483. Prerequisite pointer invoked at proof line(s) 6474; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05W5: sources/stacks-chow/tags.txt, source lines 7574–7574. Prerequisite pointer invoked at proof line(s) 6479; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Z \to Y$ be morphisms of algebraic spaces. Let $T \subset |X|$ be closed. Assume that*

(1) *X is locally Noetherian,*
(2) *g is a monomorphism and locally of finite type,*

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(3) $f|_{X \setminus T} : X \setminus T \to Y$ *factors through* g, *and*
(4) $f_{/T} : X_{/T} \to Y$ *factors through* g,

then f *factors through* g.

Proof. Consider the fibre product $E = X \times_Y Z \to X$. By assumption the open immersion $X \setminus T \to X$ factors through E and any morphism $\varphi : X' \to X$ with $|\varphi|(|X'|) \subset T$ factors through E as well, see Formal Spaces, Section 0AIX. By More on Morphisms of Spaces, Lemma 0APQ this implies that $E \to X$ is étale at every point of E mapping to a point of T. Hence $E \to X$ is an étale monomorphism, hence an open immersion (Morphisms of Spaces, Lemma 05W5). Then it follows that $E = X$ since our assumptions imply that $|X| = |E|$. □